

Enhancing Resilience of Reconfigurable Power-Water Systems with Mobile Distributed Generators and High-Proportional Renewables

Yesen Yang

Institute of Catastrophe Risk Management
Interdisciplinary Graduate Programme
Nanyang Technological University
Singapore
yesen001@e.ntu.edu.sg

Zhengmao Li

School of Electrical Engineering
Aalto University
Espoo, Finland
zhengmao.li@aalto.fi

Guangxiao Zhang

Institute of Catastrophe Risk
Management
Nanyang Technological University
Singapore
guangxiao.zhang@ntu.edu.sg

Alberto Costa

Singapore-ETH centre
ETH Zurich
Singapore
alberto.costa@sec.ethz.ch

Edmond Y. Lo*

School of Civil and Environmental
Engineering
Nanyang Technological University
Singapore
cymlo@ntu.edu.sg

*Corresponding author

Abstract—The intertwined interdependencies existing in power-water systems (PWS) increase the risk of cascading failures during post-interruption scenarios and affect the overall resiliency. Towards a more resilient operation of damaged PWS, this paper presents a resilience enhancement methodology to improve serviceability with mobile distributed generators (MDGs) and high-proportional renewables (HPR). First, the PWS is comprehensively modeled with component mechanisms and flow constraints. The interdependencies, including the power needs for water components, are modeled at component level. Second, various resources, such as extensively installed solar HPR units and MDGs, are coordinated to supply damaged PWS and reduce unsupplied loads. Reconfigurability of power distribution networks is incorporated in the proposed method. It is to adjust PDN topology by coordinating the behavior of switches and leverage HPR and MDG for optimally allocating energy. Third, the model and enhancement measures are formulated into mixed-inter linear programming to facilitate efficient solving. The developed method is applied to a benchmark PWS with 33 power buses and 25 water nodes. The simulation results demonstrate the effectiveness of our method.

Index Terms—Resilience, power-water system, reconfiguration, high-proportional renewables, mobile distributed generators

I. Introduction

Power and water systems are responsible for energy and water supply, which reliable operation is paramount for human societies. Nowadays, the accelerating urbanization and wide installation of newer technologies make the two systems increasingly interdependent which is referred to as power-water systems (PWS) [1]. The interdependence comprises power needs from water components and coupled demands from consumers. It implies the operation of one system can impact the other, and creates the

possibilities of cascading faults [2]. Common examples are disruptions in the power system that cascade to water systems by interrupting the power supply of water components and shutting the water supply. Nowadays, the rising frequency of natural hazards and extreme weather events motivates the research of system resilience. It is well-accepted that this requires evaluation of a system to resist, absorb, and recover from disruptions. In the context of PWS, enhancing resilience is to maintain the serviceability of electricity and water to the greatest possible extent [3].

Traditionally, reportedly efforts on resilience enhancements are separately conducted for power and water systems. As for power distribution systems (PDS), Ref. [4] studied load restoration using electrical vehicles as mobile grid-support resources. A multi-step strategy is proposed to integrate power flow and mobile distributed generators (MDGs) to maximize the cumulated energy service. Research in [5] combined network reconfiguration and optimal power flow to enhance the resilience of a damaged PDS. The method first adjusts the PDS topology by switching actions and then optimizes the energy dispatch for the reconfigured network. In [6], pre- and post-disruption measures are considered to improve PDS resilience, comprising placement of distributed generators, network reconfiguration, and power dispatch of energy resources. The IEEE 33-bus case system is employed to validate the proposed strategy. From [4]–[6], it can be seen that reconfiguring PDS topology, dispatch of power flow, and selective load shedding are commonly exploited for resilient PDS operation. However, the function of HPR in enhancing PDS resilience is insufficiently studied. Due to its distributed feature, HPR has the potential to provide

This research is supported by the National Research Foundation Singapore (NRF).

energy close to consumers and is suitable to synergize with reconfiguration and MDG units.

On the other hand, the inherent assumption in [4]–[6] is that power loads are always independent, i.e., they are not influenced by other factors outside PDS. Nonetheless, the power supply can change the operational state of water distribution systems (WDS). Then the coupling from unmet water demand can lead to unmet power consumption since some activities cannot proceed if they need supply from both [7]. However, the interdependencies between PDS and WDS resilience are rarely considered.

For WDS resilience, the enhancement measures involve the operation of water facilities, and dispatching of water flow. For example, Ref. [8] applies valve manipulation and pressure management to mitigate the negative impact of WDS failures. In [9], operations of tanks and pumps are exploited for minimizing the unmet water demands. However, both references assumed the power availability as a constant limitation neglecting the effects of PDS resilience. This illustrates the necessity of an integrated model to improve PWS resilience by combining consideration of damaged states and operation details of both systems.

In addition to the modeling of PWS, the uncertainties in resilience enhancement also need attention, especially the generation capacities of HPR units. General methods to deal with uncertainties comprise stochastic programming (SP) and robust optimization (RO) [10]. SP is to optimize the expectation of the objective with the probability distribution of uncertainties. It requires intensive computation and is considered over-optimistic. In contrast, RO focuses on the worst-case of uncertain scenarios [11]. It provides more robust solutions and fits the resilience study.

Given the research issues discussed above, the contribution of this paper is summarized as follows. (1) An integrated model is developed for PWS resilience enhancement. The model combines operation details of both PDS and WDS components as well as interdependencies. (2) MDG and HPR units are exploited to facilitate the resilient operation of PWS. Various enhancement measures, such as placement of MDGs, network reconfiguration, components operation, and flow dispatching are comprehensively mobilized to minimize unsupplied power and water loads. (3) The problem is originally represented with nonlinear and nonconvex constraints, and it is then reformulated through linearization and convexification techniques. The reformulated model is a mixed-integer linear programming (MILP) problem that can be readily solved with commercial solvers.

II. Model of power-water system

A general structure of PWS is shown in Fig. 1. The entire system comprises a radial power distribution network and a meshed pipe distribution network as well as various components. In PDS, the power supply comes from the main grid through the substation, while considerable additional supplies are distributionally installed, such as renewable generations and distributed generators (DG).

WDS produces potable water using treatment plants, pressurizes the water through pump stations, and then delivers the water through the pipe network. Water tanks serve as storage units in WDS and provide flexibility in balancing water production and consumption.

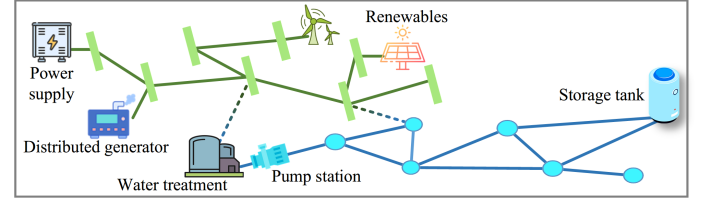


Fig. 1. A general structure of PWS.

A. Water distribution system model

The WDS network is formed with pipes and nodes. The water balance at each node is given in (1).

$$Q_{tr}^{t,i} = F_{tk}^{t,i} - R_{tk}^{t,i} + W_{rs}^{t,i} + \sum_{(i,j) \in E_w} Q_{wf}^{t,ij}, \forall i \in N_w \quad (1)$$

where i and ij are indices of nodes (N_w) and pipes (E_w). The terms respectively represent water production from the treatment plant ($Q_{tr}^{t,i}$), withdrawn water ($F_{tk}^{t,i}$) and released water ($R_{tk}^{t,i}$) of the tank, supplied water ($W_{rs}^{t,i}$) and water flow through pipes ($Q_{wf}^{t,ij}$).

In WDS, water transport is driven by the water head, which is similar to voltages in PDS and comprises pressure and elevation, as in (2). Head loss exists in pipes due to friction. It is calculated with the Hazen-Williams equation, as given in (3).

$$H^{t,i} = P_{hy}^{t,i} + h^i, \forall i \in N_w, \quad (2)$$

$H^{t,i} - H^{t,j} + H_{pu}^{t,ij} = C_{hw}^{ij} \text{Sign}(Q_{wf}^{t,ij}) |Q_{wf}^{t,ij}|^\theta, \forall (i,j) \in E_w \quad (3)$ where $H^{t,i}$, $P_{hy}^{t,i}$, h^i are water head, pressure, and elevation. $H_{pu}^{t,ij}$ is head gain from pump. The right-hand side is head loss calculation with the Hazen-Williams equation. Coefficient $C_{hw}^{ij} = \kappa L^{ij} / [(C^{ij})^\theta (d^{ij})^\mu]$ is calculated with length (L^{ij}), roughness (C^{ij}), and diameter (d^{ij}) of the pipe. $\text{Sign}(\ast)$ and $|\ast|$ are to take the sign and absolute value of " $\ast$ ". θ, κ, μ are constants.

Pumps are used to pressurize the water and maintain the water head. The operation is modeled in (4).

$$\eta_{pu}^{ij} \Gamma_{pu}^{t,ij} = \rho g H_{pu}^{t,ij} Q_{wf}^{t,ij}, \forall (i,j) \in E_w \quad (4)$$

where $\Gamma_{pu}^{t,ij}$ and η_{pu}^{ij} are consumed power and power efficiency of the pump. ρ and g indicate water density and acceleration of gravity.

Tanks experience filling and releasing water, while the change of volume is indicated in (5).

$$V_{tk}^{t,i} = V_{tk}^{t-1,i} + (F_{tk}^{t,i} - R_{tk}^{t,i}) \Delta T, \forall i \in N_w \quad (5)$$

where $V_{tk}^{t,i}$ is tank volume, and ΔT is unit operation interval.

Water treatment plant consumes energy and processes the raw water into drinking water. The energy consumption is given in (6).

$$\Gamma_{tr}^{t,i} = r_{tr}^i Q_{tr}^{t,i}, \forall i \in N_w \quad (6)$$

where $\Gamma_{tr}^{t,i}$ is power consumption of water treatment and r_{tr}^i is corresponding coefficient for energy use.

B. Power distribution system model

PDS distributes electricity from generation units to multiple consumers through power flows. DistFlow is a typical method to model power flow in PDS and is used here. The active and reactive power balances are given in (7) and (8).

$$P_{dg}^{t,i} + P_{re}^{t,i} - P_{rs}^{t,i} = \sum_{(i,j) \in \mathbf{E}_p} P_{pf}^{t,ij} + \Gamma_{pu}^{t,ij} \Lambda_{pu} + \Gamma_{tr}^{t,i} \Lambda_{tr}, \forall i \in \mathbf{N}_p \quad (7)$$

$$Q_{dg}^{t,i} + Q_{wd}^{t,i} + Q_{pv}^{t,i} - Q_{rs}^{t,i} = \sum_{(i,j) \in \mathbf{E}_p} Q_{pf}^{t,ij}, \forall i \in \mathbf{N}_p \quad (8)$$

where, i and (i, j) indicate indices for bus set $\mathbf{N}_p$ and branch set $\mathbf{E}_p$. P and Q represent active and reactive power respectively. $P_{dg}^{t,i}/Q_{dg}^{t,i}$, $P_{re}^{t,i}/Q_{re}^{t,i}$ and $P_{rs}^{t,i}/Q_{rs}^{t,i}$ are DGs' power, renewables' generation, as well as power consumption. $P_{pf}^{t,ij}/Q_{pf}^{t,ij}$ show power flows. Λ_{pu} and Λ_{tr} indicate the relation that which buses supply power to the pump and treatment plant.

The calculation for buses' voltages is formulated in (9). It holds for connected branches while relaxes for disconnected branches.

$$V_{ol}^{t,i} - V_{ol}^{t,j} = \frac{r^{ij} P_{pf}^{t,ij} + x^{ij} Q_{pf}^{t,ij}}{V_{ol}^0}, \text{ if } \pi^{ij} = 1, \forall (i, j) \in \mathbf{E}_p \quad (9)$$

where, $V_{ol}^{t,i}$ is bus voltage, r^{ij} and x^{ij} are resistance and reactance of the branch. V_{ol}^0 is reference voltage. π^{ij} indicates the binary status of branches that "1" for connection, and "0" for disconnection.

DGs consume stock fuels to generate power. The operation is formulated in (10).

$$E_{dg}^{t,i} = E_{dg}^{t-1,i} - P_{dg}^{t,i} \Delta T, \forall i \in \mathbf{N}_p \quad (10)$$

where $E_{dg}^{t,i}$ represents the amount of remaining fuels.

C. Discussion on interdependencies within PWS

As seen in the formulations in previous sections, the direct interdependencies are in power support to water components, such as pumps ($\Gamma_{pu}^{t,ij}$) and water treatments ($\Gamma_{tr}^{t,i}$). However, the indirect interdependencies existing on the consumer side also need attention. This refers to the occasions that for some activities needing both power and water, if the water supply is not available, then such activities can not proceed, and are not able to consume power. In this case, the reduction of power load is not because of consumers' intentions but constrained by unmet water demands [12]. The unmet demand coming from this reason should also be counted. Further, this relation is bidirectional and can also cascade from unmet power demands due to water losses.

Herein, the constraint from power to water supply is formulated in (11). It constrains that if the proportion of supplied power drops to $1/(1 + \varpi^{ij})$, then the proportion of water supply will be limited. A corresponding constraint from water to power is in (12).

$$W_{rs}^{t,i}/W_{dm}^{t,i} \leq (1 + \varpi^{ij}) \cdot P_{rs}^{t,j}/P_{dm}^{t,j} \cdot \Lambda_{pw} \quad (11)$$

$$P_{rs}^{t,j}/P_{dm}^{t,j} \leq (1 + \xi^{ij}) \cdot W_{rs}^{t,i}/W_{dm}^{t,i} \cdot \Lambda_{pw} \quad (12)$$

where, $W_{rs}^{t,i}$ and $P_{dm}^{t,i}$ are water/power demands. ϖ^{ij} and ξ^{ij} are positive constant coefficients. Λ_{pw} defines the relation that node i and bus j serve same consumers.

III. Resilience enhancement methods

A. Resilience curve

Following the definition of resilience, the resilience curve for PWS is shown schematically in Fig. 2.

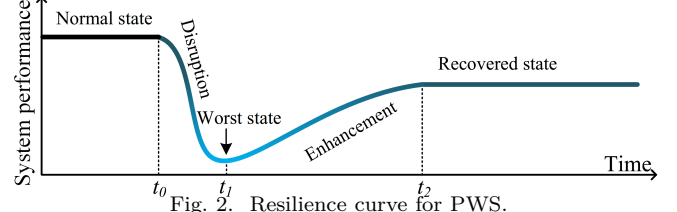


Fig. 2. Resilience curve for PWS.

It can be seen that after the disruption, the system performance quickly drops from the normal state to the worst state (t_0 - t_1). Then, resilience enhancement measures take effect to improve the performance to the greatest possible extent (t_1 - t_2). After that, PWS operates in a recovered state (t_2 -). It is necessary to note that the recovered state will likely be lower than the pre-disruption normal state since repair is not modeled here within the "short" time horizon.

To minimize the not-supplied power and water loads, the exploited enhancement measures comprise placements of MDGs, network reconfiguration, dispatch of renewables, and components' operations. The details are discussed in the following sections.

B. Placements of MDGs

MDG placement is to optimally select the locations (which bus to connect) for MDGs. These decisions directly shape the power flow and impact the network reconfiguration. Eq. (13) indicates that every MDG is assigned to connect one bus. Eq. (14) constrains that at most one MDG can be assigned to each bus. The placement results are applied to limit the generation ability of buses, as in (15).

$$\sum_{i \in \mathbf{N}_p} \beta^{i,m} = 1, \forall m \in \mathbf{M}_p \quad (13)$$

$$\sum_{m \in \mathbf{M}_p} \beta^{i,m} \leq 1, \forall i \in \mathbf{N}_p \quad (14)$$

$$P_{dg}^{t,i}, Q_{dg}^{t,i}, E_{dg}^{t,i} \leq \begin{cases} P_{dg}^{\max,m}, Q_{dg}^{\max,m}, E_{dg}^{\max,m}, & \text{if } \beta^{i,m} = 1 \\ 0, 0, 0, & \text{if } \beta^{i,m} = 0 \end{cases} \quad (15)$$

where, $m \in \mathbf{M}_p$ are index and set for MDGs. $\beta^{i,m}$ are binary variables indicating whether to place MDG m at bus i (if 1) or not (if 0).

C. Network reconfiguration

The spanning tree method is popularly used to reconfigure the topology of PDS and form a few radial islands. For each island, the bus connected to MDG is assigned as the root bus, as in (16). Eqs. (17) and (18) constrain the number of connected branches. The parent-child relation for buses in each island is given in (19).

$$\sum_{(i,j) \in \mathbf{E}_p} \lambda^{ij} = 1 - \sum_{m \in \mathbf{M}_p} \beta^{i,m}, \forall i \in \mathbf{N}_p \quad (16)$$

$$\sum_{(i,j) \in \mathbf{E}_p} \pi^{ij} = 2([\mathbf{N}_p] - [\mathbf{M}_p]) \quad (17)$$

$$\pi^{ij} = \pi^{ji}, \quad \forall (i, j) \in \mathbf{E}_p \quad (18)$$

$$\lambda^{ij} + \lambda^{ji} = \pi^{ij}, \quad \forall (i, j) \in \mathbf{E}_p \quad (19)$$

where, $[\ast]$ counts the size of set “ $\ast$ ”. λ^{ij} is binary and indicates parent-child relation for bus i - j .

IV. Problem formulation

A. Objective function

The objective is to minimize the not-supplied loads in both WDS and PDS. A weighted summation with monetary costs is utilized, as shown in (20).

$$\min \sum_{t \in T} \left\{ \sum_{i \in \mathbf{N}_w} l_w^{t,i} (W_{dm}^{t,i} - W_{rs}^{t,i}) + \sum_{j \in \mathbf{N}_p} l_p^{t,j} (P_{dm}^{t,j} - P_{rs}^{t,j}) \right\} \Delta T \quad (20)$$

where, $l_w^{t,i}$ and $l_p^{t,j}$ are assumed monetary costs for not-supplied water and power.

B. Constraints

The constraints for the proposed resilience enhancement comprise the PWS model in (1)-(12), and resilience enhancement measures in (13)-(19). The remaining operation constraints are given as follows.

$$P_{hy}^{\min,i}, V_{tk}^{\min,i} \leq P_{hy}^{t,i}, V_{tk}^{t,i} \leq P_{hy}^{\max,i}, V_{tk}^{\max,i}, \forall i \in \mathbf{N}_w \quad (21)$$

$$H_{pu}^{\min,ij}, \Gamma_{pu}^{\min,ij} \leq H_{pu}^{t,ij}, \Gamma_{pu}^{t,ij} \leq H_{pu}^{\max,ij}, \Gamma_{pu}^{\max,ij}, \forall (i, j) \in \mathbf{E}_w \quad (22)$$

$$-\pi^{ij} S_{\max}^{ij} \leq P_{pf}^{t,ij}, Q_{pf}^{t,ij} \leq \pi^{ij} S_{\max}^{ij}, \forall (i, j) \in \mathbf{E}_p \quad (23)$$

$$-\sqrt{2}\pi^{ij} S_{\max}^{ij} \leq P_{pf}^{t,ij} + Q_{pf}^{t,ij}, P_{pf}^{t,ij} - Q_{pf}^{t,ij} \leq \sqrt{2}\pi^{ij} S_{\max}^{ij}, \quad \forall (i, j) \in \mathbf{E}_p \quad (24)$$

$$0 \leq P_{rs}^{t,i}, P_{re}^{t,i} \leq P_{dm}^{t,i}, P_{RE}^{t,i}, \forall i \in \mathbf{N}_p \quad (25)$$

$$Q_{rs}^{t,i} = P_{rs}^{t,i} \tan(\cos^{-1} \varphi_{rs}^{t,i}), \forall i \in \mathbf{N}_p \quad (26)$$

$$Q_{re}^{t,i} = P_{re}^{t,i} \tan(\cos^{-1} \varphi_{re}^{t,i}), \forall i \in \mathbf{N}_p \quad (27)$$

$$V_{ol}^{\min,i} \leq V_{ol}^{t,i} \leq V_{ol}^{\max,i}, \forall i \in \mathbf{N}_p \quad (28)$$

where, $S_{\max}^{ij}$ is the maximum capacity for transmission lines in PDS. $P_{RE}^{t,i}$ is the maximum capacity for renewable generations. $\varphi_{rs}^{t,i}$ and $\varphi_{re}^{t,i}$ are power factors for power loads and renewables.

C. Model linearization and convexification

The formulated resilience enhancement problem contains several nonlinearities, including the Hazen-Williams equation in (3), pump operation in (4), and voltage calculation in (9). To facilitate efficient solving, these nonlinearities are linearized as follows.

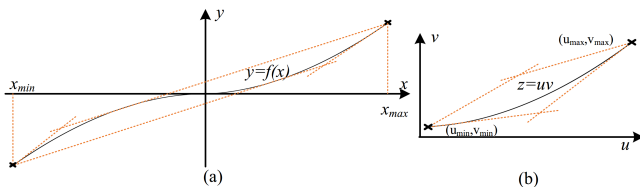


Fig. 3. Profile of (a) quasi-convex hull and (b) McCormick envelope for linearizations.

First, we relax Eq. (3) with a quasi-convex hull as given in (29) and shown in Fig. 3 (a). Readers can refer to Ref. [13] for more details.

$$\begin{aligned} & H^{t,i} - H^{t,j} + H_{pu}^{t,ij} \\ & \begin{cases} \leq (2\sqrt{2} - 2)C_{hw}^{ij} Q_{wf}^{\max,ij} Q_{wf}^{t,ij} + (3\sqrt{2} - 2)C_{hw}^{ij} (Q_{wf}^{\max,ij})^\theta \\ \geq (2\sqrt{2} - 2)C_{hw}^{ij} Q_{wf}^{\min,ij} Q_{wf}^{t,ij} - (3\sqrt{2} - 2)C_{hw}^{ij} (Q_{wf}^{\min,ij})^\theta \\ \geq 2C_{hw}^{ij} Q_{wf}^{\max,ij} Q_{wf}^{t,ij} - C_{hw}^{ij} (Q_{wf}^{\max,ij})^\theta \\ \leq 2C_{hw}^{ij} Q_{wf}^{\min,ij} Q_{wf}^{t,ij} - C_{hw}^{ij} (Q_{wf}^{\min,ij})^\theta \end{cases} \end{aligned} \quad (29)$$

McCormick envelope is utilized to relax the variable production in (4), as shown in (30) and Fig. 3 (b).

$$\begin{aligned} & \eta_{pu}^{ij} \Gamma_{pu}^{t,ij} / (\rho g) \\ & \begin{cases} \geq H_{pu}^{\max,ij} Q_{wf}^{t,ij} + H_{pu}^{t,ij} Q_{wf}^{\max,ij} - H_{pu}^{\max,ij} Q_{wf}^{\max,ij} \\ \geq H_{pu}^{\min,ij} Q_{wf}^{t,ij} + H_{pu}^{t,ij} Q_{wf}^{\min,ij} - H_{pu}^{\min,ij} Q_{wf}^{\min,ij} \\ \leq H_{pu}^{\max,ij} Q_{wf}^{t,ij} + H_{pu}^{t,ij} Q_{wf}^{\min,ij} - H_{pu}^{\max,ij} Q_{wf}^{\min,ij} \\ \leq H_{pu}^{\min,ij} Q_{wf}^{t,ij} + H_{pu}^{t,ij} Q_{wf}^{\max,ij} - H_{pu}^{\min,ij} Q_{wf}^{\max,ij} \end{cases} \end{aligned} \quad (30)$$

The binary states in (9) are linearized with a big scatter, M , as in (31).

$$-M(1 - \pi^{ij}) \leq V_{ol}^{t,i} - V_{ol}^{t,j} - \frac{r^{ij} P_{pf}^{t,ij} + x^{ij} Q_{pf}^{t,ij}}{V_{ol}^0} \leq M(1 - \pi^{ij}) \quad (31)$$

After the linearization and convexification, the problem is reformulated into a MILP problem, which can be readily solved by off-the-shelf solvers.

D. Discussion on handling HPR uncertainties

In the formulated problem, the main uncertainty comes from the generation capacity of renewables, i.e., $P_{RE}^{t,i}$. It is also observed that this parameter only appears in (25) and works as an upper bound of exploited energy of renewables ($P_{re}^{t,i}$). Hence, a robust decision can be made by taking the lower boundary values of the estimated renewables' generation and letting them constrain the exploited power from renewables.

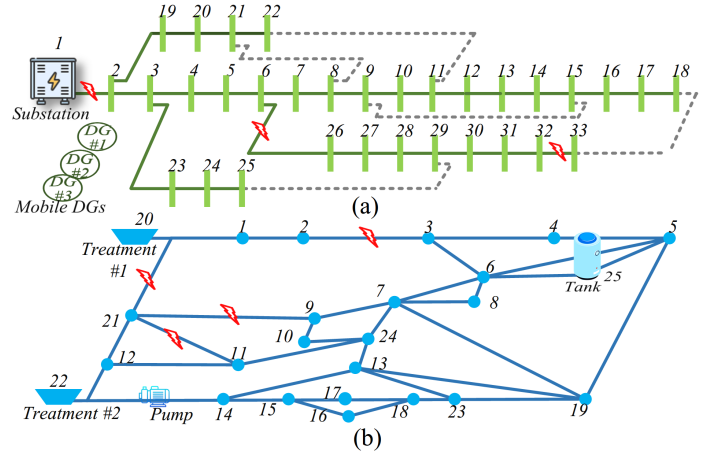


Fig. 4. Profile of (a) 33-bus and (b) 25-node PWS system

V. Case study

A. Setting of test PWS system

The 33-bus/25-node PWS system is utilized to demonstrate the effectiveness of the proposed resilience enhancement method. The profile is given in Fig. 4. In PDS, five tie-lines (marked with grey dash lines and normally open) are exploited to support network reconfiguration.

HPR is distributionally installed in buses, with capacities indicated in Table. I [14]. The profile of renewable generations is given in Fig. 5. Three MDGs with operation parameters in Table II are available, where the initial energy is set as the maximum. In WDS, two treatment plants, one pump station, and one tank are included, with operation parameters shown in Table. III [15]. The power supply to water components is from bus #23 to treatment #1, bus #31 to treatment #2, and bus #4 to the pump. The overall demands profile is given in Fig. 6. The initial volume of the tank is set as 60% of the maximum volume. The bus-node pairs regarding interdependent demands (in Eqs. (11), (12)) are (#5-#10), (#7-#7), (#11-#8), (#13-#6), and (#14-#4). The parameters, ϖ/ξ are set as 0.2 and 0.5. The time horizon is 24 hours with a one-hour operation interval.

TABLE I
Location and capacities of HPR units

Bus (#)	3	8	17	22	26	30	33
Capacity (kW)	100	80	80	40	40	100	100

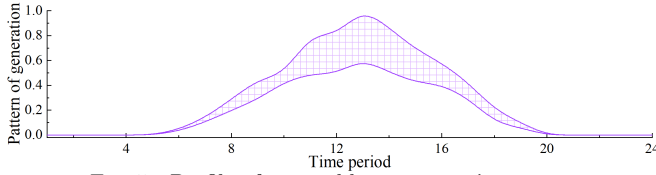


Fig. 5. Profile of renewable generation's pattern

TABLE II
Operation parameters for MDGs (units: kW, kVar, kWh)

$P_{dq}^{\max,1}$	$Q_{dq}^{\max,1}$	$E_{dq}^{\max,1}$	$P_{dq}^{\max,2}$	$Q_{dq}^{\max,2}$	$E_{dq}^{\max,2}$	$P_{dq}^{\max,3}$	$Q_{dq}^{\max,3}$	$E_{dq}^{\max,3}$
1350	1280	1730	1350	1280	1304	1350	1280	1220

TABLE III
Operation parameters for water facilities

$r_{tr}^{t20}/r_{tr}^{t22}$	η_{pu}^{t22t4}	ρ	g	$V_{tk}^{\max,25}$
1 kWh/m ³	0.85	1000kg/m ³	9.8m/s ²	1200m ³

After a disruption, three power lines and four pipes are assumed to be broken (marked with red lightning). The resilience enhancement works to minimize the unsupplied loads, with the monetary costs shown in Fig. 7. The case study is conducted with an AMD 5800H 3.20GHz PC with 32GB RAM and solved by the Python/Gurobi. The convergence criteria is set at 0.1% change in the objective (20).

B. Numerical results

With the available MDGs and renewable generations, the program works to find the optimal enhancement decisions.

First, the placements for three MDGs are (MDG #1 at bus #23), (MDG #2 at bus #31), and (MDG #3 at bus #4). Fig. 8 shows the decisions on the network reconfiguration for PDS. The observations based on Fig. 8 are concluded as follows.

- (1) After the reconfiguration, three power islands are formed with MDGs being root buses for each island.

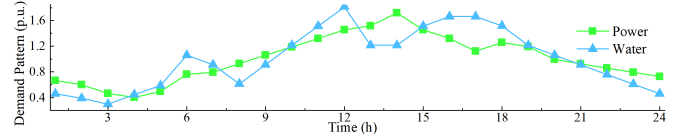


Fig. 6. Profile of power and water demands' patterns

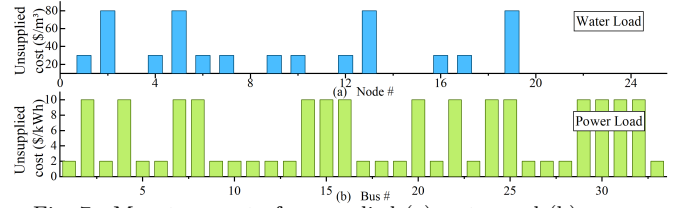


Fig. 7. Monetary cost of unsupplied (a) water and (b) power

- (2) All three islanded subnetworks maintain the radial topology.
- (3) The island #3 is the largest island with the placed MDG #3 having the lowest reserve fuel but complemented with the majority of HPR units included in the island.

All these observations show the effectiveness of the proposed enhancement method.

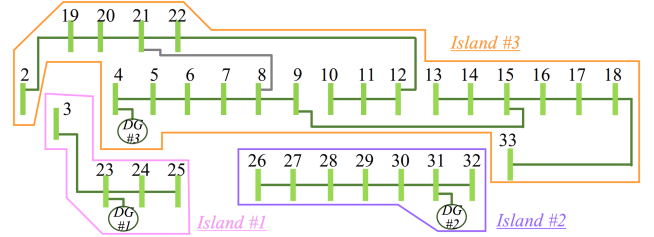


Fig. 8. Profile of power islands after placement of MDGs and network reconfiguration

The fuel usage of three MDGs is shown in Fig. 9. It can be seen that all the MDGs gradually consume the reserved fuels until depletion. It demonstrates the rationality of assigning loads to MDGs through location selection and reconfiguration.

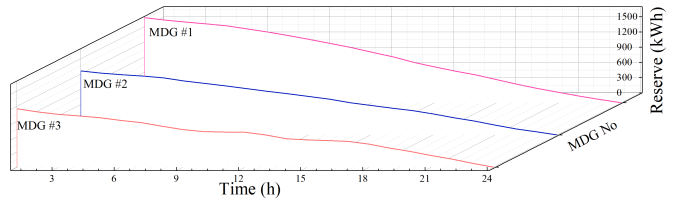


Fig. 9. Reserved energy for three MDGs

The operations of the two treatment plants and the tank in WDS are shown in Fig. 10 and 11. The observations are as follows.

- (1) The production of treatment #1 is limited due to pipe breakages.
- (2) The treatment #2 always works on the maximum producing state, with full power support from PDS.
- (3) The tank releases water for the water supply and also withdraws water from the WDS for some time, which helps keep treatment #2 from reducing supply for balancing demands and maximizes the water production.

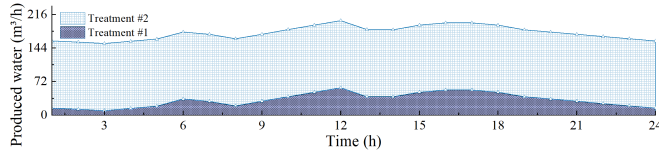


Fig. 10. Profile of water production of treatment plants

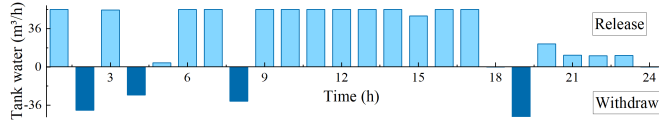


Fig. 11. Operation of the tank in WDS

Fig. 12 (a) and (b) show the overall power and water supply. It can be seen that for WDS, the produced water is distributed to nodes based on their importance, i.e., interruption costs. Similar trends appear in PDS while more “less important” loads are also satisfied when abundant renewable energy is available (around the 14th hour). This indicates the new configuration can fully accommodate renewable generations. The total monetary cost is \$374,926 and the computing time is 71.9s.

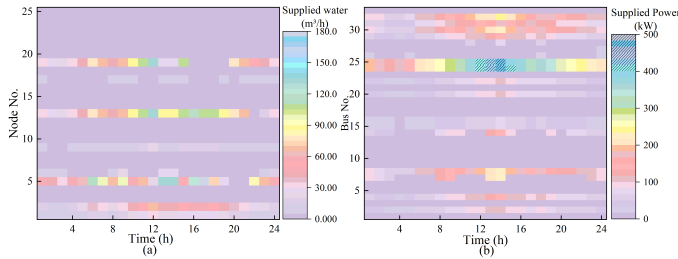


Fig. 12. (a) Supplied water to nodes and (b) supplied power to buses

C. Comparison with conventional enhancement methods

Several conventional methods are used for comparison and to further validate the effectiveness of the proposed resilience enhancement strategy.

- M1: Individual enhancement of PDS and WDS with placement of MDGs and network reconfiguration;
- M2: Joint enhancement with MDGs' placement only and fixed topology (no reconfiguration).
- M3: Joint enhancement with network reconfiguration only and fixed DGs' location.

The results of simulations are listed in Table IV.

TABLE IV
Monetary costs for all methods

M1	M2	M3	Proposed method
\$715,499	\$605,746	\$443,626	\$374,926

From the result, the comparison with M1 shows the benefits of combining consideration of PDS and WDS resilience, especially for maintaining WDS operation. The comparison with M2 indicates it is vital to conduct network reconfiguration since it helps reform power islands and pick up disconnected loads. The comparison with M3 indicates the advantage of location selection for MDGs that it can better assign the energy resources to loads.

VI. Conclusion

In this paper, we propose a resilience enhancement method for interdependent PWS with MDGs and renew-

able generation. With the problem formulation and case study, it can be concluded that:

- (1) The proposed enhancement method is efficient in improving the resilience of both PDS and WDS;
- (2) Considering interdependencies, the joint model of PWS can be superior to individual PDS and WDS models in resilience studies;
- (3) A combination of MDG placement and network reconfiguration can further optimize the power flow with limited energy and accommodate renewable generations from HPR units.
- (4) Linearization and convexification techniques support the efficient solving of PWS resilience problems.

References

- [1] A. M. Hamiche et al., “A review of the water-energy nexus,” *Renewable and Sustainable Energy Reviews*, vol. 65, pp. 319–331, 2016.
- [2] F. Wang et al., “Modeling cascading failure of interdependent critical infrastructure systems using HLA-based co-simulation,” *Autom. Constr.*, vol. 133, 2022.
- [3] M. S. Roni et al., “Integrated water-power system resiliency quantification, challenge and opportunity,” *Energy Strategy Reviews*, vol. 39, p. 100796, 2022.
- [4] AK. Erenoglu et al., “Resiliency-Driven Multi-Step Critical Load Restoration Strategy Integrating On-Call Electric Vehicle Fleet Management Services,” *IEEE Trans. SMART GRID*, vol. 13, no. 4, pp. 3118–3132, 2022.
- [5] YL. Li et al., “Restoration Strategy for Active Distribution Systems Considering Endogenous Uncertainty in Cold Load Pickup,” *IEEE Trans. SMART GRID*, vol. 13, no. 4, pp. 2690–2702, 2022.
- [6] M. Haghsheenas et al., “Power distribution system resilience enhancement planning against extreme dust storms via pre- and post-event actions considering uncertainties,” *Sustainable Cities and Society*, vol. 78, p. 103626, 2022.
- [7] G. Y. Qiu et al., “A quantitative study on the water-related energy use in the urban water system of Shenzhen,” *Sustainable Cities and Society*, vol. 80, p. 103786, 2022.
- [8] H. A. Mahmoud et al., “Real-Time Operational Response Methodology for Reducing Failure Impacts in Water Distribution Systems,” *J. Water Resour. Plann. Manage.*, vol. 144, no. 7, p. 04018029, 2018.
- [9] P. Khatavkar et al., “Resilience of Water Distribution Systems during Real-Time Operations under Limited Water and/or Energy Availability Conditions,” *J. Water Resour. Plann. Manage.*, vol. 145, no. 10, p. 04019045, 2019.
- [10] Z. Li et al., “Restoration of Multi Energy Distribution Systems With Joint District Network Recon Figuration by a Distributed Stochastic Programming Approach,” *IEEE Trans. Smart Grid*, pp. 1–1, 2023.
- [11] A. Costa et al., “An optimization model for power grid fortification to maximize attack immunity,” *International Journal of Electrical Power & Energy Systems*, vol. 99, pp. 594–602, 2018.
- [12] Y. Yang et al., “Risk-averse restoration of coupled power and water systems with small pumped-hydro storage and stochastic rooftop renewables,” *Appl. Energy*, vol. 339, p. 120953, 2023.
- [13] Q. Li et al., “Micro Water-Energy Nexus: Optimal Demand-Side Management and Quasi-Convex Hull Relaxation,” *IEEE Trans. Control Netw. Syst.*, vol. 6, no. 4, pp. 1313–1322, 2019.
- [14] Z. Li et al., “Coordinated preparation and recovery of a post-disaster Multi-energy distribution system considering thermal inertia and diverse uncertainties,” *Applied Energy*, vol. 336, p. 120736, 2023.
- [15] F. Pecci et al., “Scalable Pareto set generation for multiobjective co-design problems in water distribution networks: A continuous relaxation approach,” *Struct. Multidisc. Optim.*, vol. 55, no. 3, pp. 857–869, 2017.